Exercise 1: Happy Supplies Parts Order

Original table

CustomerNumber	CustomerName	CustomerType	Date	Time	Employee	PartNumber	Name	Type	Cage Code	Quantity Ordered	Unit Price
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10654	Float Control	Plumbing	G413	4	4
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

1(a). Assumptions

1. CustomerNumber uniquely identifies each customer, and CustomerName and CustomerType depend on CustomerNumber.

2. PartNumber uniquely identifies each part, and Name, Type, CageCode, and UnitPrice depend on PartNumber.

3. An order is identified by CustomerNumber, Date, and Time because the form does not provide an order number.

4. Each order is associated with one employee, and employees can serve any customer.

5. An order can contain multiple parts, and QuantityOrdered depends on the order and PartNumber.

1(b). Normalization process

1. UNF: The original form contains a repeating group of ordered parts. The relation is PartsOrder(CustomerNumber, CustomerName, CustomerType, Date, Time, Employee, {PartNumber, Name, Type, CageCode, QuantityOrdered, UnitPrice}).

2. 1NF: Each ordered part is placed on a separate row so every attribute contains one value. The primary key is (CustomerNumber, Date, Time, PartNumber). The relation is PartsOrder_1NF(**CustomerNumber**, **Date**, **Time**, **PartNumber**, CustomerName, CustomerType, Employee, Name, Type, CageCode, QuantityOrdered, UnitPrice).

3. 2NF: CustomerName and CustomerType depend only on CustomerNumber, while Name, Type, CageCode, and UnitPrice depend only on PartNumber. These partial dependencies are removed by separating the data into Customer_2NF(**CustomerNumber**, CustomerName, CustomerType), Part_2NF(**PartNumber**, Name, Type, CageCode, UnitPrice), OrderHeader_2NF(**CustomerNumber**, **Date**, **Time**, Employee), and OrderLine_2NF(**CustomerNumber**, **Date**, **Time**, **PartNumber**, QuantityOrdered).

4. 3NF: Under the stated assumptions, the 2NF relations have no transitive dependencies, so they also satisfy 3NF. The final relations are Customer(**CustomerNumber**, CustomerName, CustomerType), Part(**PartNumber**, Name, Type, CageCode, UnitPrice), OrderHeader(**CustomerNumber**, **Date**, **Time**, Employee), and OrderLine(**CustomerNumber**, **Date**, **Time**, **PartNumber**, QuantityOrdered).

1(c). Primary and foreign keys

1. In 1NF, the primary key is (CustomerNumber, Date, Time, PartNumber).

2. In 2NF, the primary keys are CustomerNumber for Customer_2NF, PartNumber for Part_2NF, (CustomerNumber, Date, Time) for OrderHeader_2NF, and (CustomerNumber, Date, Time, PartNumber) for OrderLine_2NF.

3. In 3NF, Customer has primary key CustomerNumber, and Part has primary key PartNumber. OrderHeader has primary key (CustomerNumber, Date, Time), and CustomerNumber is a foreign key referencing Customer. OrderLine has primary key (CustomerNumber, Date, Time, PartNumber); (CustomerNumber, Date, Time) is a foreign key referencing OrderHeader, and PartNumber is a foreign key referencing Part.

1(d). The named 3NF relations are Customer, Part, OrderHeader, and OrderLine.

Exercise 2: Panacea Mental Health Corporation

Original table

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

2(a). Assumptions

1. staffNo uniquely identifies each therapist and determines therapistName; patNo uniquely identifies each patient and determines patName.

2. An appointment is identified by (patNo, appointmentDate, appointmentTime), assuming a patient cannot have two appointments at the exact same time.

3. Each appointment has one therapist and one branch. A therapist may work at different branches on different dates but only one branch on a given date.

2(b). Normalization process

1. UNF: The original relation is TherapyAppointment(staffNo, therapistName, patNo, patName, appointmentDate, appointmentTime, branchNo).

2. 1NF: The table contains atomic values, and the primary key is (patNo, appointmentDate, appointmentTime). The relation is TherapyAppointment_1NF(**patNo**, **appointmentDate**, **appointmentTime**, staffNo, therapistName, patName, branchNo).

3. 2NF: Patient and therapist details are separated because patName depends on patNo and therapistName depends on staffNo. Branch assignments are stored by therapist and date. The relations are Patient_2NF(**patNo**, patName), Therapist_2NF(**staffNo**, therapistName), TherapistWorkday_2NF(**staffNo**, **appointmentDate**, branchNo), and Appointment_2NF(**patNo**, **appointmentDate**, **appointmentTime**, staffNo).

4. 3NF: The final relations are Patient(**patNo**, patName), Therapist(**staffNo**, therapistName), TherapistWorkday(**staffNo**, **appointmentDate**, branchNo), and Appointment(**patNo**, **appointmentDate**, **appointmentTime**, *staffNo*). BranchNo is stored once per therapist and date rather than repeated for each appointment.

2(c). Primary and foreign keys

1. In 1NF, the primary key is (patNo, appointmentDate, appointmentTime).

2. In 2NF, the primary keys are patNo for Patient_2NF, staffNo for Therapist_2NF, (staffNo, appointmentDate) for TherapistWorkday_2NF, and (patNo, appointmentDate, appointmentTime) for Appointment_2NF.

3. In 3NF, Patient has primary key patNo, and Therapist has primary key staffNo. TherapistWorkday has primary key (staffNo, appointmentDate), with staffNo as a foreign key referencing Therapist. Appointment has primary key (patNo, appointmentDate, appointmentTime), with patNo referencing Patient, staffNo referencing Therapist, and (staffNo, appointmentDate) referencing TherapistWorkday.

2(d). The named 3NF relations are Patient, Therapist, TherapistWorkday, and Appointment.

Exercise 3: Maid Better Temp Agency

Original table

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

3(a). Assumptions

1. eNo uniquely identifies each employee and determines eName.

2. contractNo uniquely identifies each contract, and each contract applies to one event.

3. eventNo identifies each event and determines eventLoc.

4. An employee can work on multiple contracts, and a contract can involve multiple employees. hours depends on the combination of eNo and contractNo.

3(b). Normalization process

1. UNF: The original relation is EmployeeContract(eNo, contractNo, hours, eName, eventNo, eventLoc).

2. 1NF: All attributes contain atomic values, and the composite primary key is (eNo, contractNo). The relation is EmployeeContract_1NF(**eNo**, **contractNo**, hours, eName, eventNo, eventLoc).

3. 2NF: eName depends only on eNo, while eventNo and eventLoc depend on contractNo. The relations are Employee_2NF(**eNo**, eName), Contract_2NF(**contractNo**, eventNo, eventLoc), and EmployeeContract_2NF(**eNo**, **contractNo**, hours).

4. 3NF: contractNo determines eventNo, and eventNo determines eventLoc, so eventLoc is transitively dependent on contractNo. The final relations are Employee(**eNo**, eName), Event(**eventNo**, eventLoc), Contract(**contractNo**, *eventNo*), and EmployeeContract(**eNo**, **contractNo**, hours).

3(c). Primary and foreign keys

1. In 1NF, the primary key is (eNo, contractNo).

2. In 2NF, the primary keys are eNo for Employee_2NF, contractNo for Contract_2NF, and (eNo, contractNo) for EmployeeContract_2NF.

3. In 3NF, Employee has primary key eNo, Event has primary key eventNo, and Contract has primary key contractNo with eventNo as a foreign key referencing Event. EmployeeContract has primary key (eNo, contractNo), with eNo referencing Employee and contractNo referencing Contract.

3(d). The named 3NF relations are Employee, Event, Contract, and EmployeeContract.